

Reinforcement Learning using EEG signals for Therapeutic Use of Music in Emotion Management

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Abstract—Prolonged influence of negative emotions can result in clinical depression or anxiety, and while many prescribed techniques exist, music therapy approaches, coupled with psychotherapy, have shown to help lower depressive symptoms, supplementing traditional treatment approaches. Identifying the appropriate choice of music, therefore, is of utmost importance. Selecting appropriate playlists, however, are challenged by user feedback that may inadvertently select songs that amplify the negative effects. Therefore, this work uses electroencephalogram (EEG) that automatically identifies the emotional impact of music and trains a reinforcement-learning approach to identify an adaptive personalized playlist of music to lead to improved emotional states. This work uses data from 32 users, collected in the publicly available DEAP dataset, to select songs for users that guide them towards joyful emotional states. Using a domain-specific reward-shaping function, a Q-learning agent is able to correctly guide a majority of users to the target emotional states, represented in a common emotion wheel. The average angular error of all users is 57° , with a standard deviation of 2.8 and the target emotional state is achieved.

Clinical relevance— Music therapy for improving clinical depression and anxiety can be supplemented by additional emotion-guided music decisions in remote and personal settings by using automated techniques to capture emotional state and identify music that best guides users to target joyful states.

I. INTRODUCTION

Disruptive emotional homeostasis is associated with fragile mental health [1]. Prolonged influence of negative emotions eventually alters the neurological circuitry of the brain and gives rise to ailments such as clinical depression or anxiety [2]; however, prescribed medications for these disorders may have various side-effects [3]. Music, as a medium of emotion regulation, may supplement traditional treatment methods and have positive impacts on anxiety levels, heart rates, and blood pressure[4][5]. Music therapy, coupled with psychotherapy, can lead to statistically significant lower depressive symptoms [6].

The choice of appropriate music is important to ensure positive outcomes from music-based intervention [7]. Music selection is important, especially in individuals vulnerable for depressive symptoms, and these individuals may not be able to select appropriate music for healthy emotional transitions [8]. Indeed, individuals with feelings of despondency can misuse music as unhealthy coping mechanisms like emotion focused coping [9], rumination [10], or social withdrawal

[11]; depressed listeners are naturally drawn to sad music despite its potentially-negative impact on their cognitive state [10]. Therefore, a solution to predict or derive the optimized sequence of music, which can drive the current emotional state of a subject to the desired cognitive state is desired.

Reinforcement learning-based approaches to generate personalised playlists have aimed to address this goal of optimal sequence of music. However, these approaches traditionally incorporate user feedback (pauses, replays, and skips of songs) [12]. The publicly available DEAP dataset recorded electroencephalogram (EEG) and peripheral physiological signals of 32 participants while each of them was watching 40 one-minute long excerpts of music clips [13], which provides a wearable sensors-based approach to automatically identifying user preference for a reinforcement learning agent. Indeed, classifying emotional signals with the help of EEG data has been a focus of recent [14]. However, blindly adapting such data to Q-learning agents requires specialized reward functions or reward-shaping techniques to help guide learning agents to avoid local minimum solutions that would impact performance. Such reward-shaping is necessary because the policy space is too large to explore exhaustively [15] and reinforcement learning often needs domain-specific reward shaping to optimize performance [16]. This work aims to use music and emotion data made available from DEAP, extracted from EEG signals rather than using the data from direct user feedback, to train a Q-learning agent to automate the personalized selection of music sequences using a specialized reward-shaping function.

II. DATA SET DESCRIPTION

We used the DEAP dataset [13] for our implementation, which recorded electroencephalogram (EEG) and peripheral physiological signals of 32 participants. while each watched 40 one-minute long excerpts of music clips. For each clip, a fixation screen was displayed and a 2-minute long EEG signal was recorded to calibrate each user to a relaxed state prior to viewing a music clip. The EEG data can be processed using algorithms such as support vector machines (SVM), long short-term memory networks (LSTM), k nearest neighbor (KNN), etc.[14] to obtain individual valence, arousal, dominance and preference values; the DEAP dataset contains processed valence and arousal, which is used in this work.

III. METHOD

A. Music recommendation using Reinforcement Learning

We employed a Q-learning model for developing the music recommendation system. The reward considered was based

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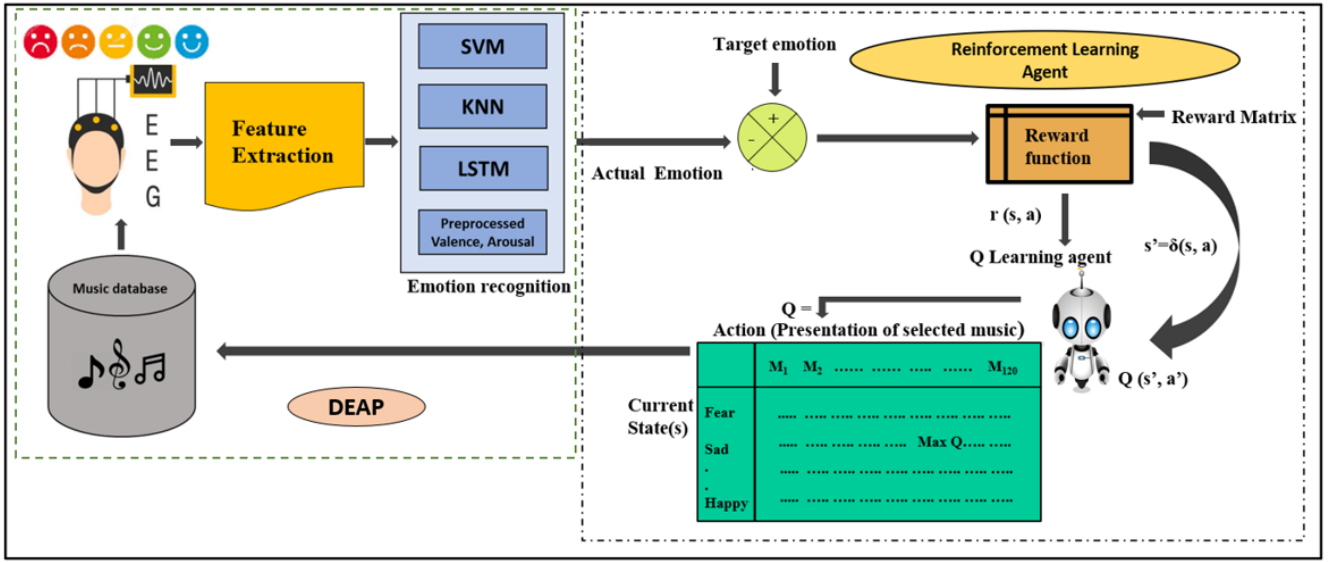


Fig. 1: Our proposed music recommendation system trained with Q-learning reinforcement learning.

on the emotions as mapped in the 2D Geneva Emotion Wheel (GEW), where positive emotions are in the upper right quadrant, negative/depressive emotions in the lower left quadrant, anger in the upper left quadrant, and hopefulness in the lower right quadrant [17]. The GEW defines a 2D circular space with horizontal axis representing valence (positive or negative influence of an emotion) and vertical axis representing arousal (the degree of excitement or calmness associated with the current cognitive state). An action represents presentation of a selected music to a given subject to influence his/her emotions and the entire model is illustrated in Fig. 1.

To model the emotion transition, we begin with training the Q-learning agent which is represented in the form of a Q-table. The adaptation rule used to iteratively update the Q table for a current emotional state s , action a and next emotional state s' is given in the standard Q-learning equation:

$$Q(s, a) = (1 - \alpha)Q(s, a) + \alpha[r(s, a) + \gamma \max_{a'} Q(s', a') + \phi(s, s')]$$

where $s' = \delta(s, a)$ is the next state due to action a at state s , $r(s, a)$ is the reward obtained by agent for taking action a at state s , $\phi(s, s')$ is the transition function due to reward shaping and α, γ are scale factors in $[0, 1]$.

During the control phase, we determine the current emotion of the subject by his/her EEG data. Next, the learning agent refers back to the trained Q-table and selects the best possible music clip to drive the subject towards targeted mental state.

B. Intermediate State Representation

To determine the impact of the music clip on the current cognitive state of the individual, we use the GEW and a (valence, arousal) tuple for each clip and each subject. Thus, the impact of each music clip on a subject can also

be envisioned as a vector on the 2-D plane as described above, and transitions as angular rotations between these coordinates. The intermediate mental state of a subject post listening to a music clip can thus be determined by the transition function denoted by $\tau(s, a)$ such that

$$\tau(s, a) = \vec{v}_s + \vec{v}_a, \vec{v}_s = \tau(s, a)$$

where $\vec{v}_s$ is the potential intermediate next state emotion vector, s is the current emotional state represented as $\vec{v}_s$ in GEW, and a represents the subjected music inducing emotion denoted by $\vec{v}_a$. This transition function is motivated by the idea that the mental state of an individual subjected to an external stimulus would depend on the nature of the stimulus and the initial emotional state of the individual. Iterative adaptation of the transition function helps us define the next state table which captures the potential emotional states corresponding to the responses to music stimuli from a given cognitive state.

C. Reward Function

In this model, the basic reward function provided an agent an immediate reward of +1 if the presented action drove the cognition closer to the target emotional state and 0 otherwise. The reward at state s (current emotional state) for an action a (music clip) is denoted as:

$$r(s, a) = \begin{cases} 1, & \text{if } \theta(s, a) \leq 180^\circ \\ 0, & \text{otherwise} \end{cases}$$

where, $\theta(s, a) = |\theta_1| + |\theta_2|$ and

$$\theta_* = \arccos \left(\frac{\vec{v}_s \cdot \vec{v}_a}{|\vec{v}_s| |\vec{v}_a|} \right)$$

where θ_1 uses $\vec{v}_t$ and θ_2 uses $\vec{v}_s$, with $-\pi < \theta_1, \theta_2 \leq \pi$. The motivation for the aforesaid reward function is to

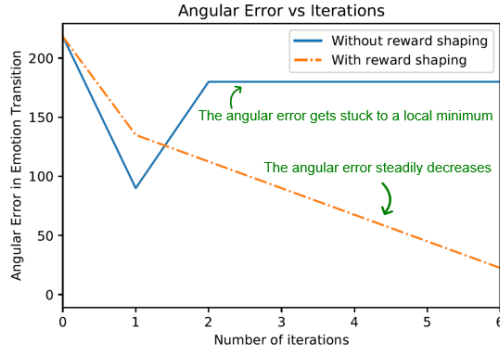


Fig. 2: Angular Error (degrees) and reward shaping.

encourage the agent to select music stimuli in such a way that the subject is driven towards the target emotional state.

D. Reward shaping

Building a robust emotion control model requires an exhaustive music library and user reaction to this library in order, which can invoke all possible emotion transitions. Due to this limitation of building the action space (and the data available in DEAP all stemming from a neutral state), the Q-learning agent often greedily selected the locally optimal, which was not the targeted emotion. For example, the Q-learning agent suggested a music clip that put a subject in a state of shame, and no music clips in the database exist that brought the user out of this state. This was mitigated by introducing prior domain knowledge to train the agent so that it was guided towards the globally optimal solution by shaping the reward function.

We introduce an additional reward for a successful state transition. That is, not only was the agent rewarded for making a transition towards a more positive emotion but also was rewarded for making any transition from one state to another, irrespective of it being positive/negative. This ensures that the agent never gets stuck to any cognitive state, other than the target positive emotional state. The shaping function for transitioning from current state (say, s) to the next state (say, s') is defined as:

$$\phi(s, s') = \begin{cases} \lambda, & \text{if both } s \text{ and } s' \text{ are same} \\ 0, & \text{otherwise} \end{cases}$$

As we wanted to discourage the agent from remaining in the same emotional state, λ assumed a large negative value for practical implementations, -100 (a tenth of regret value).

IV. EXPERIMENT AND RESULTS

Using the extracted emotion data from the EEG readings in the DEAP dataset, this work conducted a leave-one-subject-out cross-validation to train a Q-learning agent and evaluate the ability of the agent to provide a playlist that would result in the targeted emotional state (*happy, pride, anger, disgust, guilt, sadness, interest, and relief*). We reduced this from the 16 emotions on the wheel by splitting every missing

TABLE I: Mean angular error (and standard deviation) for Total (T) 32 subjects, the 19 subjects that achieve target emotional state success (S), and remaining 13 subjects (F) over six music clips (iterations)

	Clip 1-2	3	4	5	6
T	68.3(14.1)	67.4(10.9)	62.9(9.9)	60.1(11.5)	56.9(11.1)
S	54.7(3.6)	39.1(3.2)	38.3(3.1)	23.1(2.4)	11.3 (2.7)
F	88.2(17.9)	108.0(16.8)	99.1(15.3)	114.1(18.1)	123.7 (17.3)



Fig. 3: Mean angular error & standard error

emotion into two, merging into neighboring emotions, partially due to label imbalance on the removed emotions. These experiments, along with the necessary hyperparameter tuning and corresponding results, are explained in the subsequent sections.

A. Hyper-parameter Tuning

The Q-Learning model had three hyper-parameters that needed to be tuned: learning rate (α), discount factor (γ), and exploration/exploitation ratio (ϵ). These hyper-parameter were tuned using grid search method with an internal 10-fold cross-validation on the training data, resulting in optimal vales of α as 0.1 and γ as 0.6. The model was also evaluated for the value of ϵ as 0 and 0.1. It was observed that in both cases, the regret (integral of angular error) increased linearly for increasing iterations initially. However, as iterations progressed, the regret tended to saturate with ϵ as 0.1 but was unbounded in the other scenario. This suggested that the agent learnt the problem better while adopting an ϵ -greedy learning approach.

B. Model Evaluation

Initially, when the Q-table was trained without incorporating reward shaping, the Q learning agent was unable to suggest a sequence of music clips to drive the target state to happy. This is evident from Fig. 2 where the increasing and subsequent saturation of angular error (without reward shaping) indicated that the model failed to converge to the target emotion.

Alternatively, Fig. 2 demonstrates that the Q learning agent was able to suggest sequences when the Q-table was trained with reward shaping. Of the 32 subjects, 19 cases converged to the target happy state. Fig. 3 lists the average (mean) angular error (with standard deviations) over all participants, and Table I breaks further into those that achieved the target

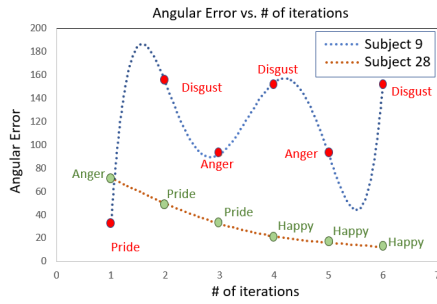


Fig. 4: Angular error and emotional state for two subjects: 9 who converged to the target state and 28 who did not

emotional state and those that did not, over six iterations. Importantly, for those 19 users a total of 9 clips are selected, demonstrating that while the first clips may be standard, unique playlists were selected based upon each subject's EEG data captured during the music clips as the iterations increased. The Q learning agent was able to create a playlist that consistently led users towards the happy target state, and the more music clips allowed the more users converged. The angular error in the sixth iteration converged to less than 60° . The number of iterations was obtained by empirical analysis and increasing the iterations further did not substantially improve the results.

This is likely a function of two findings: 1) that the number of similar joyous targets, and an imbalance in the selection of joyous states in the DEAP dataset, allow most users to get close to our designated target of happy; and 2) the larger error bars illustrate that for 19 subjects the angular error approaches 0 as each individual reaches the key target state, and those that do not converge have a wider array of possible angular error values. Two such cases (one convergent, one divergent) are illustrated in Fig. 4. The average angular error in the sixth iteration for the 19 participants is 11.3° , with a standard deviation of 2.8, while the average angular error in the sixth iteration for the remaining 13 participants is 123.7° , with a standard deviation of 17.3. EEG data is able to find the target state for 19 users and the Q learning agent is able to do so with a minimal number of iterations. Further exploration is needed to determine why subject 28 in Fig. 4, and similar such subjects, oscillate between emotional states but do not converge on the target, perhaps because of the reward shaping mechanism employed to aid most subjects.

V. LIMITATION AND FUTURE WORKS

We create personalized playlists to help subjects arrive at target emotional states. We can explore additional rewards based upon additional input data including further processing of EEG data and acquiring user feedback, which may demonstrate variations along the emotion wheel. We will develop a sequence of music to fine-tune the emotional experiences and demonstrate effectiveness across emotional states. Ultimately, randomized control trials are needed to validate the effectiveness in improving emotional states.

VI. CONCLUSIONS

This work developed a reinforcement learning approach for developing music playlists targeted at emotion regulation based upon wearable EEG data. Using just six music clips in a playlist, 19 of the 32 subjects in the publicly available DEAP dataset arrived at a joyful target emotional state, with a mean angular error of 11.3° (standard deviation: 2.8°), with the overall mean angular error of 57.0° , when calculated in a leave-one-subject-out cross-validation. Importantly, this work demonstrated an angular error-based reward shaping function that a Q-learning agent used to select songs to arrive at a target emotion, even if each individual clip would not. This work presented a framework of taking processed, wearable EEG data to derive personalized, emotion-driven music playlists, which enables future studies on emotional health applications and interventions.

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